Gears, motor and first power

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Compare a gear-based prediction with a supervised first powered test.

SUCCESS CRITERIA - I can...

1. Check gear mesh and secure connections before power.
2. Calculate ideal wheel speed from measured or supplied rpm.
3. Document first-test evidence and one revision.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

At unchanged wheel rpm, circumference becomes 3 times as large. Compare ideal travel speed and state one real-world limitation.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: rotation rate = turns/Δt
v = circumference × rotation rate.
Turns per second, no slip. Get first-power approval; secure car and keep loose items clear. Disconnect before adjusting gears.

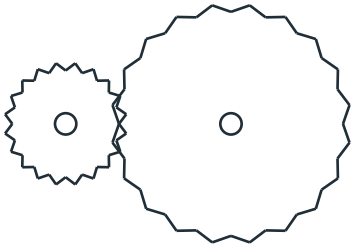
Worked reasoning: Supplied wheel tests show 4 rotations in 2 s. With circumference 0.20 m and no slip, predicted speed is(4/2)×0.20=0.40 m/s. Ground speed may differ if wheels slip.

Your turn: Draw one wheel rotation and the predicted travel distance. Identify the no-slip assumption.

Marzano check - circle: ① ② ③

03 Look and explain

Sketch your driver and driven gears. Label tooth counts and identify the driven axle. Note any rubbing or loose mesh before power.



Driver Driven

Marzano check - circle: ① ② ③

04 Build the solution

A supplied motor turns 300 rpm through a 10-tooth driver and 30-tooth driven gear. Use the supplied wheel rate 100 rpm and circumference .18 m. Find ideal speed=.18×100÷60.

Marzano check - circle: ① ② ③

05 Use the idea

The actual car is slower than the prediction. Name two assumptions to check and one measurement that could separate their effects.

Marzano check - circle: 1 2 3

06 Choose a method

A consistently drawn magnetic-field map has 4 times as many field lines through the same small perpendicular area at A as at B. What does the denser spacing indicate?

Marzano check - circle: 1 2 3

07 Exit check

Record teacher approval, first-test observation and one next action.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A car's wheels predict 0.40 m/s but measured ground speed is 0.30 m/s. Name two hypotheses and a measurement to distinguish them.

Marzano check - circle: 1 2 3